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The Arabic Language, Arabic Linguistics and Arabic Computational Linguistics

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1 Introduction

Arabic is a natural language and as such, shares many properties with other natural languages. Firstly, Arabic is primarily a VSO language which is the third most common word order among the world's languages (Broadwell, 2005). It also has a relatively free word order and like Italian and Spanish it is a pro-drop language. The building blocks of Arabic words are nonconcatentative morphemes (McCarthy, 1981) such as the consonantal root and the vocalic melody. And similar to other agglutinative languages, it has a rich and complex word structure. Constituent question phrases are usually fronted in Arabic as they are in French, English and German, with the exception of Egyptian Arabic (Farghaly, 1981). Like Hebrew and many other languages, Arabic retains a resumptive pronoun in the formation of some relative clauses. Through these shared properties, interesting insights can be gained (Aoun and Li, 2003, Farghaly, 1981, Fehri, 1993, Benmamoun, 2000) by analyzing the structure of Arabic by any of the prevailing linguistic theories or formalisms (Chomsky, 1965, 1981, 1995, Bresnan, 2000, Sag and Wasow, 1999). In section 2.4 we will discuss in detail the relevance of the Arabic language for the advancement and refinement of linguistic theories.

Although Arabic shares many properties with other languages, it also has some features that distinguish it and that merit further research. For example it is very unusual for a language like Classical Arabic (CA) to persist as a living and functional language for almost 1500 years although it is not the mother tongue of anyone (Holt, 1995). The stable sociolinguistic situation that has accommodated the coexistence of both classical Arabic and dialects poses interesting questions. And no other language, with the exception of Hebrew, exhibits such rich morphological derivation as the majority of the words are formed from a limited number of roots.

In this chapter we touch upon the similarities that Arabic shares with other languages and focus on the features of Arabic that differentiate it from other languages. In the

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next section of this chapter, a brief survey of the Arabic language is presented, highlighting the importance and relevance of the study of the Arabic language from historical, cultural, strategic and linguistic perspectives. The focus in section 3 is on the Arabic linguistics tradition beginning in the 8th century AD. We present an account of the great achievements of this tradition as well as its limitations. Then we discuss current work in Arabic linguistics with respect to the dilemma facing some Arab linguists; that is, balancing the need for innovation in analyzing Arabic with the pressure to conform to and preserve the early Arabic linguistic tradition. In section 4, we relate the recent achievements in Arabic Computational Linguistics that are reported in this book with the history of Arabic Computational Linguistics which began in the late 1970's and early 1980's. Section 5 concludes the chapter.

2 The Arabic Language

The Arabic language is at present the most widely spoken Semitic language with the number of Arabic speakers in the Middle East approaching 330 million according to the CIA (CIA, 2008). It is the sixth most widely spoken language in the world and one of the six official languages of the United Nations.

The Arabic language is considered the defining property of Arabic identity (Suleiman, 1994) and there is no other language that so uniquely defines the identity of its own people. Arabic speakers in the Middle East do not originate from a single ethnic background nor do they belong to one religion. The Syrians and Lebanese have a Phoenician origin while some Egyptians consider themselves the descendants of the Pharaohs. Many Iraqis are of Kurdish origin while many Moroccans, Algerians and Tunisians are Berber. The majority of the Saudis and the peoples of the Gulf states are the offspring of Bedouins, oasis-dwelling farmers, and fishermen. There is a minority Assyrian population living in Iraq.

In addition to the multiplicity of ethnic backgrounds, the Arab countries are also characterized by a plurality of religions. The land of the Arab countries is the homeland of the three monotheistic religions: Judaism, Christianity and Islam. It has also witnessed the splits and formation of different sects within these main religions such as Copts, Maronites, Orthodox, Catholics, Shiites, Sunnis, Wahabiyyiin, Druze, Alawiyyiin, etc.

However, the most significant factor that unites all these people is that they are all native speakers of one form or another of Arabic.

2.1 Historically

The history of the Arabic language is interesting and has important theoretical implications especially for the origin of Arabic diglossia (Ferguson, 1959b) and for language stability (Holt, 1995). The Arabic language evolved from an obscure, non-prestigious language to a major world language after the Islamic conquests. And although Classical Arabic was standardized, codified and fixed as early as in the eighth century A.D., there is still a controversy over when, where and how the present Arabic dialects evolved. Unlike other languages, Classical Arabic, throughout its history, adapted itself to meet the changing needs of its speakers. Thus, it evolved from primarily a poetic language in the pre-Islamic period to the language of science, mathematics, chemistry, astronomy and

medicine as well as the language of government and religion. Yet, there were periods when the prestige and functionality of Classical Arabic were demoted during the five centuries when the Arabs came under the rule of the Ottoman Caliphs. So, there must be an explanation for the stability of Classical Arabic which has lasted for over fifteen centuries even though it is not the native language of living group.

2.1.1 The First Standardization of Arabic

Until the 7th century AD, the world knew very little about the Arabic people or the Arabic language (Versteegh, 1997a). Before Islam, Arabic was only a minor member of the southern branch of the Semitic language family. It was spoken primarily in the Arabian peninsula and only by small groups in the wider area outside the peninsula as a minority language because Aramaic was the 'lingua franca' in the region. With the emergence of Islam, the Arabs emerged from the Arabian Peninsula to call other nations to convert to Islam and within fifty years they succeeded in controlling the entire Middle East region and within a hundred years, they reached the Atlantic Ocean via North Africa and Spain. With the creation of the Islamic Empire, Arabic became a major world language and the world could no longer afford to ignore the Arabic language nor the Arab people.

Prior to the emergence of Islam, Arabic was spoken as a native language in the Arabian Peninsula and Yemen and was a minor language among the other Semitic languages such as Babylonian, Assyrian, Canaanite, and Aramaic (Versteegh, 1997a). However, native speakers of Arabic at that time, who were primarily nomads and Bedouins, spoke different dialects of Arabic. The dialect spoken by a tribe called Quraish قَبيلَةُ قُرَيش *qabylatu qurayš* in Mecca is assumed to have been the most prestigious Arabic dialect due to the elevated position the tribe held among other Arab tribes and due to its wealth and strategic location in Mecca. Mecca at that time was a trade and religious center which housed all the idols worshiped by the Arabs and all Arab tribes would make a pilgrimage to Mecca to worship their gods. Most significantly, the Prophet Mohammed was a member of the Quraish tribe and it was their dialect that became the language of the Quran.

The first codification of the Arabic language was undertaken by early Arab grammarians in the 8th century who regarded the language of the Quran as the model of correctness. This was the first time the Arabic language was standardized with an explicit grammar defining correct usage. The codification included all its linguistic levels such as its phonology, morphology, syntax and lexicons. So for over 1500 years and to the present, this grammar of classical Arabic is still taught to all Arabic speakers in their general education courses. Following the expansion of the Islamic State, the Arabic language became the official language of administration, government and religious teaching in the new empire. Thus, Classical Arabic satisfied all the conditions (Mesthrie et al., 2000) that modern linguistics prescribes for a standardized language. Firstly, it had a codified grammar and dictionaries that define the norms of the languages. Secondly, it was prestigious because it was the language of the Quran and was also assumed to be the language of God. Thirdly, it was a model representing the correct and "pure" form of Arabic. Fourth, it functioned as the language of government, administration, and

education. Fifth, although it was associated with a powerful and prestigious subgroup (Quraish) it was also the language that new converts to Islam needed to learn in order to perform their religious rituals. Thus, the function of Classical Arabic extended beyond the communicative needs of its native speakers (Mesthrie et al., 2000) as it served as the standard that all Arabic speakers aspire to master.

2.1.2 The Emergence of the Dialects

The widely accepted Western view is that there were three Arabic language varieties in the pre-Islamic period عصرُ الجَاهِلِيةِ *ṣru 'lǧāhilyti* and early Islamic era. They were:

1. The language of the Quran which Muslims believed to be the language of God, and as such, was a perfect language and superior to any other language
2. The poetic language which exhibited the Arabs fondness for oratory. Arabs from various tribes would compete to create the most flowery poetic language which was highly rhythmical and had elaborated sixteen meters.
3. Everyday language was the language before Islam that Arabs used in their daily interaction. There is no consensus on how proximate this variety was to the language of the Quran or to the poetic language.

In contrast, most Arab linguists believe there was only one language. The Arab view is that the difference between the language of the Quran and the Poetic language was not that significant. Both are highly elevated forms of Arabic with rhythmical and rhetorical patterns. With respect to "everyday language", people most likely spoke a language similar to that recorded in the Quran. This is most probable because as a Muslim, you believed the Quran was a gift from God to the people of the Arabian peninsula and must be in the language of the people living there. So naturally it must be in their everyday language as the Quran was not directed toward the educated elite, but rather, the common people. On the other hand, if you were not a Muslim you might believe that the Quran was written by the prophet himself. So it must be written in a language he knew and spoke and that was shared by the people around him. So, in either case, the Quranic language was not different from the everyday language spoken at the time of the Prophet.

Shortly after the death of the prophet Mohammad in 632 CE, many Arabs left the Arabian Peninsula to live in the different territories under the Islamic rule which extended from the Arabian Peninsula to the Pacific Ocean. Two processes were taking place simultaneously in the centuries that followed. Versteegh (1997a) calls them Arabicisation and Islamisation. Arabicisation proceeded more rapidly than Islamisation due to the "prevailing tolerance on the part of the Muslims towards Christians and Jews" which did not generate an urgent need to convert to Islam" (Versteegh, 1997a, pg. 93). It was language, rather than religion, that became the binding force in the Islamic Empire.

The indigenous populations of the conquered countries greatly outnumbered the native Arabs and they spoke different languages. For example the Iraqis spoke Aramaic, the Syrians Greek and the Egyptians Coptic, but there was a need for linguistic accommodation, mostly on the part of the conquered so that they could communicate with their conquerors about matters of taxation, administration and trade (Versteegh, 1997a). Further, learning and mastering Arabic would provide employment opportuni-

ties for the local people. The early attempts made by the locals at speaking Arabic were tainted by the traits of their native languages and there have been numerous reports of the frequently unsuccessful attempts by indigenous peoples to communicate with their rulers. The new speakers of Arabic spoke the language and read the Quran with an accent. Early Arab grammarian's efforts to codify Arabic were motivated in part, by the fear that the purity of the Arabic language would be tainted by the "corrupt" Arabic spoken by the local people لحنُ العَامَةِ *lḥnu 'l-ʿāmati.* Arab grammarians felt they needed to write rules to clearly define the "correct" ways of speaking Arabic so their goals were prescriptive in the sense of both controlling the behavior and the manner in which Arabic would be used. Once the language was codified, all speakers had to follow these rules.

Just as there are differing views on the number of early Arabic languages, there are also different theories about the origin of modern Arabic dialects. Many linguists (Versteegh, 1997a) assume that modern Arabic dialects developed from this first colloquial Arabic spoken during the early days of the Arab Conquests and other Arabic grammarians believe that modern dialects developed from Classical Arabic. As an example, people who did not know how to speak Classical Arabic correctly, tended to drop case endings, spoke it with an accent and introduced lexical innovation. A third view is that of Ferguson (1959a) who refutes the claim that these dialects are descendants of Classical Arabic. He cites fourteen linguistic features (mostly phonological and morphological features) that all dialects share but that are lacking in Classical Arabic. He proposes that all Arabic dialects originate from a form of Arabic spoken at the camps of Arabic military bases positioned in the conquered territories.

Further, there is no agreement on how many varieties actually exist today. For example Ferguson (1959b) refers to two varieties: the high variety, or Classical Arabic and the low variety which is used in the everyday communication of Arabic speakers. Badawi (1973) cites five varieties of Arabic. But, broadly speaking, there are at least three varieties of Arabic that co-exist side by side.

1. Classical Arabic is the most prestigious variety because it is the language of the Quran. It is well defined because it has been codified by early Arab grammarians. The consensus among traditional Arabic grammarians is that this grammar is complete as it describes a closed corpus i.e., the Arabic religious and literary heritage.
2. The colloquial dialects are also well defined; not because they are fully codified, but because they are acquired naturally by their native speakers. Each Arabic speaking country has its own dialect which is used primarily in everyday communications.
3. Modern Standard Arabic (MSA) is the form of Arabic that is used among educated speakers of Arabic in formal situations. It is not a well defined variety because, unlike the colloquial dialects, it is not the native language of anyone. And unlike Classical Arabic, it has not been fully elucidated and described.

2.1.3 Arabic as the Language of Science and Philosophy

There is a near consensus that a highly elevated poetic form of Arabic existed in the pre-Islamic period but Islam introduced a new dimension to the Arabic language as it

became the language of religion. Soon after the death of the Prophet and the establishment of Islamic rule in the greater Middle East, a third dimension was added as Arabic became the language of government and administration by the 8th century.

The codification of the Arabic language in the 8th century gave birth to an intellectual movement in the Arab World. The Arabs made an effort to understand the heritage of the Roman, Greek and Indian civilizations when seeking answers to the questions their new intellectual endeavors were addressing. The Arabic language lexicon was greatly enriched and expanded with the translations of the great Greek and Roman writers. Even though during this time, the Arabs were the sole keepers of earlier civilizations, their role was not limited to simply preserving the earlier heritage. They were able to make important contributions to knowledge after they had mastered contemporary scientific knowledge. From the 10th to the 13th centuries there was a massive proliferation of books in the urban centers. Having amassed a huge body of knowledge, they produced encyclopedias such as those of Al-Khawarizmi اَلخَوَارِزمي *al-ḫawārizmy*, Al-Ghazali and Ibn Sina الغَزَالي و إِبن سينَا *al-ġazāly w ʾibn synā* and Ibn Rushd ابن رُشـد *ābn rušd* known in the west as Averroes and for his Aristotelian comments (Harvey, 2000).

Biesterfeldt (2000) describes Al-Farabi's encyclopedia إِحصَاءُ العُلومِ *ʾiḥṣāʾu ʾl-ʿulwmi* 'Enumeration of Sciences' as "having the greatest impact on the classification of sciences in Islam, and beyond, during the Christian and Jewish Middle Ages. The medieval western world came to know the classical works of the Greeks either directly through the Arabic translations or indirectly through the medieval Hebrew encyclopedias that, in turn, were largely influenced by the Arabic encyclopedias" (Biesterfeldt, 2000).

Arab scholars did not have a unified view of how to classify knowledge. For example, Al-Khawarizmi اَلخَوَارِزمي *al-ḫawārizmy* in his encyclopedia مِفتَاحُ العُلومِ *miftāḥu ʾl-ʿulwmi* 'Keys to the Sciences' classified sciences into two main categories: Islamic/Arabic Sciences and foreign sciences. He listed six branches of knowledge under Islamic/Arabic sciences including jurisprudence, theology, grammar, scribal art, poetry and prosody, and history. Under foreign sciences he refers to philosophy, logic, medicine, arithmetic, geometry, astronomy, music, mechanics and alchemy (Biesterfeldt, 2000). On the other hand Al-Farabi اَلفَارَابي *al-fārāby* classified sciences differently in his encyclopedia 'Enumeration of Sciences' إِحصَاءُ العُلومِ *ʾiḥṣāʾu ʾl-ʿulwmi.* He provided a general survey of each of the sciences in his encyclopedia which consisted of five topical chapters.

1. The Science of Language
2. The Science of Logic
3. The sciences of music, mathematics: arithmetic, geometry, optics, mathematical astronomy, the weights, and mechanics.
4. Natural Science and Divine Science (metaphysics)
5. Political Science, Jurisprudence and Theology.

The impressive work produced by the Arab scholars shortly before and after the Abbasid Caliphate proves that the Arabic language, which had been regarded as primarily a flowery and poetic language, adapted itself to the emerging needs of its speakers. Thus it became a vehicle for expressing complex accounts of scientific advances in almost all fields of knowledge. The scientific advances of the Arabs at the time raised the

functionality of Classical Arabic to its highest level. It became the language of science and philosophy in addition to being the language of religion, literature and government. It is worth noting that the The Center for Muslim Contribution to Civilization in the State of Qatar recently launched a project to translate the landmarks in Arabic scientific thought into English. The center has already published and is in the process of publishing several English translations of Arabic books written by medieval Arabic scholars.

But after the fall of the Abbasid Dynasty, the pre-eminence of the Arabic language began to decline. For the first time in the history of Islam, the caliphs were non-Arabs who were not proficient Arabic speakers. Under Turkish rule, the Arabic language was no longer the language of government. We will talk below about how the position of the Arabic language was demoted for at least four centuries.

2.1.4 The Demotion of the Arabic Language

It is important to understand the role of language in government under Islamic rule. The use of the Arabic language in government began when the Prophet Mohammad migrated from Mecca to Yathrib in 622 AD يَثْرِب *yaṯrib* which was renamed Al-Madina اَلمَدِينَةُ *al-madyntu*. Upon his arrival he drafted a set of deeds known as the Constitution of Medina (CM) (Arjomand, 2009). The CM is probably the very first constitution in either the Arabian Peninsula or the entire world and the CM clearly established the Prophet as the head of the "Umma" الآمَةُ *āl-ʾāmatu*. The CM created a political community on the basis of well-defined common interests, rights and duties. This was a significant departure from the traditional tribal rule where loyalty to the tribe was paramount. The newly created political community transcended tribal allegiance, transforming tribal allegiance to political allegiance. The new political community included Muslims and non-Muslims آهلُ الكِتَابِ *ʾāhlu ʾl-kitābi* forming a pact that clearly defined the rights and obligations of the three factions of the Umma in Medina:

1. The Muslims from Mecca who migrated to Medina اَلمُهَاجِرون *al-muhāǧirwn*
2. The newly converted Muslims in Medina who welcomed the Prophet and his followers from Mecca اَلآنصَارُ *al-ʾānṣāru*
3. The Jews in Medina

Following his death, Muhammad was succeeded by four caliphs known as the Rashiduun Caliphs الخُلَفَاءُ الرَاشِدون *al-ḫulafāʾu ʾl-rāšidwn* who originated from the Prophet's clan and were selected by consensus among Muslims. After the death of the fourth Caliph, Ali Ibn Abi Taaleb, the Caliphate was reshaped into dynasties and the caliphs within each dynasty were descendants of the same family. The Caliph in turn, would select governors وَالي *wāly* for each province and/or country under his rule and the governor's role was clearly defined: maintain peace and order within his province, follow the Islamic jurisprudence and collect taxes from Muslims اَلزَكَاةُ *al-zakātu* and also collect اَلجِزيَةُ *al-ǧizytu* taxes from non-Muslims.

Under the Umayyad Caliphates, Islamic rule expanded westward across North Africa and into Spain and eastward to the Punjab encompassing the three major continents: Asia, Africa and Europe. The Rashiduun Caliphs, the Umayyads and the subsequent

Abbasids all originated from the tribe of Quraish. So the language of Quraish enjoyed high prestige at the time. Further, all the appointed rulers of the conquered countries were Arabs. It was natural then for the Arabic language to become the official language of government in all countries under Islamic rule. All formal correspondence, instructions, judicial and legislative proceedings, were conducted in the Arabic language.

In 1258, the Abbasid caliphate collapsed after the attack of the Mongol forces who conquered Baghdad and Al-Musta'sim Billah المُعتَصِمُ بالله *āl-muʿtaṣimu b-al-lāh* the last Abbasid caliph who ruled from Baghdad, was killed in the attack. The fall of the central Islamic Caliphate in Baghdad and the death of the last Abbasid caliph marked the beginning of a different era characterized by the emergence of several small shadow caliphates and resulted in the disintegration of the Islamic empire. In Egypt for example, the Mamluks were not of Arabic origin, yet they hosted an Abbasid Caliph, and did so only to legitimize their rule. Hence, the real power was in the hands of the Mamluks while the Abbasid Caliph exerted little influence and usually capitulated to their demands.

Then more than two hundred years later in 1517, the Turkish Sultan Selim I conquered Egypt and captured the nominal Abbasid caliph, al-Mutawakkil III. He was taken to Istanbul and was forced to surrender the Caliphate to the Turks. With the fall of the Abbasid Caliphate, a new Islamic dynasty formed, known as the Osman Caliphate الخِلَافَةُ العُثمَانيةُ *āl-ḫilāfatu āl-ʿuṯmānytu.*

During this time, while Classical Arabic remained the language of religion, it lost its role as the language of government and administration. For the first time in Islamic history, the Caliph was neither an Arab nor a native speaker of Arabic. However, he was the religious leader of the empire. As a result during this period, religion became the only factor that unified the people under his leadership.

Turkish was the predominant language of the Sultans in Constantinople and more often than not, the Caliph would assign Turkish governors who did not speak Arabic fluently, to govern the Arabic provinces. This motivated the local Arab speaking population to learn Turkish in order to obtain administrative positions and to resolve issues with their rulers. The prior supremacy of the Arabic language was replaced by a multiplicity of three role-specific languages: Turkish for government, Persian for literature and Arabic for religion. The relegation of Arabic to solely the realm of religion persisted for three centuries until well into the nineteenth century which witnessed the rebirth of Arabic (Versteegh, 1997a).

Despite the setback suffered by the Arabic language during the centuries under Turkish domination, it has surprisingly, survived and regained its vitality in the nineteenth century when the Arab countries came into direct contact with the ideas, culture, philosophy and scientific and technological advances in Western Europe. As a result, this contact between the Arabs and Europeans in the nineteenth century had a profound influence on both the Arabic culture and language as discussed in the next subsection.

2.1.5 The Second Standardization of Arabic: The Emergence of Modern Standard Arabic

As stated above, the Arabic language revived itself in the nineteenth century via Modern Standard Arabic (MSA) which emerged when Western concepts had to be translated

into Arabic and many Arabists (Versteegh, 1997a) consider the emergence of MSA as a rebirth of Classical Arabic.

In 1798, Bonaparte led the French expedition across the Mediterranean and occupied Egypt until 1801. The French wanted to intercept the trade route between Great Britain and its colonies in the Far East in order to establish a base to destroy the British-Indian empire (Peretz, 1994). Hence, there was a renewed awareness on the part of the Western powers as to the strategic importance of the Middle East. For the first time, since Egypt came under Ottoman rule in 1517, it came into direct contact with the Western world. The Egyptians were taken by surprise when the French soldiers used guns which to them, were new weapons and they realized how backward they had become as compared to the Western World. Bonaparte brought a printer to Egypt able to print the Arabic script and he printed leaflets directed toward the Egyptians in order to win their "hearts and minds". Whether intentional or not, the French expedition exposed the Egyptians to the "new" Western world and its culture. Egypt became even more influenced by Western culture when in 1805 it came under the rule of Mohammad Ali Pasha. He realized that the Turkish Caliphate in Istanbul was in decline and wanted to establish a strong Islamic empire that extended beyond Egypt. To achieve his goal, he initiated an ambitious program to modernize and industrialize Egypt. He introduced a new administrative system and reorganized the fiscal and agrarian systems (Holt, 1966). He built an organized system of government with a leader "mudiir" for each function: finance, navy, war, interior, education and public works marking the beginning of the cabinet system in Egypt. Following the advice of his French advisers, he sent young Egyptian men to France to study medicine, engineering and all modern branches of science. Upon their return from France, they taught at the new technical schools or "colleges" founded by Ali Pasha to teach ship-building, medicine, surveying, etc. So as to prepare Egyptians for these technical schools, he developed the first secular state education system in Egypt. He also established a language school to facilitate the translation of technical materials into Arabic. His openness to the West, especially to France, brought to Egypt the new ideas and concepts in vogue in Western Europe at the time.

But the Arabic language had to be used to express ideas and concepts that did not exist in Classical Arabic. New Arabic words such as 'constitution', 'republic', 'representations', 'national assembly', etc. had to be created in addition to thousands of technical and scientific expressions. This was an important impetus for the development of Modern Standard Arabic which originated with the Western-educated Arabs who needed to present political, cultural and technical knowledge in Arabic. The development of MSA has continued to the present and now functions as the mutually accepted vehicle of communication among educated Arabs from different regions in the Arabic-speaking world.

2.1.6 The Stability of Classical Arabic

Classical Arabic has remained intelligible, functional and widely used throughout the last fifteen centuries and is currently the official language of twenty two states. Holt (1995) says:

> "of the four major literate civilizations: Confucianism, Hinduism, Christianity and Islam, it is only Islam whose sacred language has retained its original form and still became a national and official language. Another way of putting it is that, whereas most sociolinguistic study is concerned with variability, here we are concerned with its opposite. Furthermore, other neighboring languages have undergone major structural and orthographic reform most related to contemporary speech patterns, i.e. Greek, Turkish, Modern Hebrew and Maltese" (Holt, 1995).

So when comparing language stability, the Arabic language has remained relatively stable in contrast to Old English which became incomprehensible to speakers of Middle English which in turn became incomprehensible to speakers of Modern English. The same can be said of Latin which lost its functionality and is now considered to be a "dead language", but gave birth to the Romance languages.

Holt (1995) cites several factors that contribute to the stability of the Arabic language. Firstly and from the very beginning, Islam did not separate state and religion and according to Islamic tenets, the prophet was the religious leader of Muslims as well as their chief political leader. In fact, the prophet became the leader of the very first centralized Arab state. Prior to Islam, the Arabs' loyalty was to the tribe but with the advent of Islam, their loyalty shifted to religious and political leaders. The fact that the head of the Arabic state was both a political and religious leader protected the Arabic language from coerced change that would have been imposed by divided leadership. Secondly, the early expansion of Islam beyond the Arabian peninsula produced an Arabic "fingerprint" or Arabocentrism and an Arabic ideology (Tibi, 1990). Thirdly, literary and scientific work was conducted in Classical Arabic and the use of Classical Arabic was an essential element in preserving the link between past and present. Fourthly, the literate minority اَلعُلَمَاء *al-ʿulamāʾ* in the Muslim and the Arab world were key to maintaining social cohesion since as Imams, they had the linguistic knowledge necessary for interpreting the Quran. They preserved and adhered to the structure of Classical Arabic in order to facilitate communication and resolve debates among themselves. Fifth, even though the اَلعُلَمَاء *al-ʿulamāʾ* were a small minority, they were able to forge a consensus and preserve the structure of Classical Arabic. In fact, what is now known as the "Arab sciences" is primarily linguistic knowledge. Thus, the specialized role of the Ulamaa عُلَمَاء *ʿulamāʾ* coupled with their ideological commitment to preserve the Quran, helps explain the remarkable homogeneity of the literary genres of the Muslim world.

2.2 Strategically

The Arabic language is the official language of 22 countries occupying a strategic geographical location and their combined size is about one and half times the size of the USA. These countries extend from the Atlantic Ocean on the west to the Arabian/Persian Gulf in the east and also include a large part of Africa and a significant portion of Asia. Further, the Arabic-speaking North African countries have close ties with Southern European countries; namely, Spain, Italy, Greece and France due to their proximity and shared history.

Throughout history the Arab countries played a pivotal role as the crossroad for east-west trade. To facilitate trade from Western Europe to India, Indonesia and China,

the Egyptians dug a canal that provided for the passage of ships coming from Europe to the Far East circumventing Africa and the treacherous route around the Cape of Good Hope. The Suez canal provided a safer, faster and less expensive alternative for transporting goods from the Far East to Europe.

During the expansion of the Roman and the Byzantine empires preceding Islam, Judaism and Christianity spread among the Arabs as Semitic religions. For example the city of Yathrib يَثْرِب *yaṯrib* became Judaized while many Arabs became Christians. Small Jewish communities lived comfortably in the Arabian Peninsula and spoke both Aramaic and Arabic. The prophet Mohammed sought the support of the Jewish communities in his call to Islam. In the Quran, Muslims are urged to respect and accommodate "the people of the Book" اهلُ الكِتَابِ *āhlu 'lkitābi* referring to believers in the Holy Book. With the rise of Islam, these three monotheistic religions not only originated in close proximity to each other but they remained in close contact with each other. The geographical proximity and cultural similarities shared by the people of these three religions makes this region particularly significant.

The region, with its large and growing population, huge oil revenues, and strategic position presents attractive business opportunities for the rest of the world. First, it is a large market for many products. Second, the Arab's keen interest in acquiring and assimilating new technological innovations to further employment and improve their standard of living draws many experts from all over the world to contribute to that effort. Third, the monetary surplus generated by oil revenues is actually reinvested in the Western world and such investment is very important to the economies of the advanced countries in the West.

2.3 Culturally

Ethnographically, the Arabs were nomads who lived, as the oldest inscriptions show, (Antonius, 1969) in the area between the Euphrates and the center of the Arabian Peninsula including southern parts of Hijaz and Najd. Life in the northern part of the Arabian Peninsula was difficult due to the lack of water resources and was an inhospitable place in which to live due to desertification. Rain was scarce and there were no rivers to support agriculture. The society was formed by closely allied tribes where loyalty to the tribe was paramount. Because there was no agriculture and hence no settled communities, the nomads (also called Bedouins) eked out an existence by moving their herds from one place to another in search of scarce resources اَلكَلَأ *alkalaʾ* and water. There were frequent battles among these tribes over resources and over control of caravan lines. There were three important cultural traits that defined the life of Arab nomads in the Peninsula before Islam.

1. Love of poetry. They excelled in using the most florid, eloquent and rhythmic forms of the Arabic language and used these metaphors, similes, and imagery extensively in their poetry. Arabic poetry dates to the fifth century CE and was memorized, recited and handed down from one generation to the next. The seven most prominent Arab poems in the pre-Islamic period are called مُعَلَّقَات *muʿalaqāt* and were hung on the walls of the ka'ba in Mecca. Hence, they are named "The

Suspended Odes or The Hanging Poems". Each one has a different author and is considered to be the author's best work.

2. They valued and praised hospitality and took the utmost pride in being hospitable and generous toward guests even sacrificing their own sustenance for a hungry guest. This reverence toward guests is reflected in their poetry.
3. But bitter tribal warfare was commonplace before Islam as exemplified by the continual bloody conflicts of Aws and Khazraj, Abs and Dhubyan, Bakr and Taghlib. The Arab nomads valued courage, stamina and horsemanship and demonstrated supreme courage in battle, taking a keen sense of pride in their skills.

In addition to the nomads, there were other Arabs who lived in settled towns surrounding the few oases. One of these towns, Mecca, was as early as the 7th century, a renowned cultural and trade center and was also the birthplace of the prophet Mohammed.

2.3.1 Arabic as an Oral Culture

Arabic discourse has its roots in an oral culture (Johnstone, 1991) that dates to the pre-Islamic period. As mentioned above, pre-Islamic poetry played an important role in the social life of the people. Longer poems (odes) were often memorized and recited orally and the tradition of recitation prevailed even after the founding of Islam. In fact, the Quran was recited long before it was written and Muslims to this day recite parts of the Quran at least five times a day in their daily prayers. Poetry such as that of the Ummayads and Abbasids was publicly recited and even lengthy prose as in "The Arabian Nights" was always recited orally. Story telling was considered a profession and had a name in the Arabic culture اَلحَكَوَاتي *al-ḥakawāty* 'the story teller'.

Traditional Islamic education took place at informal schools called the katatiib كَتَاتيب *katātyb*; a tradition that continues to the present day, relying heavily on repetition and memorization. At these schools young children memorize the entire Quran at a very early age without necessarily understanding its meaning. The emphasis on repetition and memorization in the Arabic culture has led some linguists to perceive it as a lack of reasoning in Arabic discourse (Johnstone, 1991, 1983). We will elaborate on the properties of Arabic discourse in section 2.5.1 below.

2.3.2 Arabic Culture and Islam

There is an unbreakable link between the Arabic language and culture, and Islam. For example, it is difficult to translate concepts specific to Islam without the use of Arabic terms. Terms such as "Jihad" جِهَاد *ǧihād*, "Imam" إِمَام *ʾimām*, "da'awa" دعوة *dʿwt*, "Quran" قرآن *qrʾān* to name but a few can only be defined with these words. Because Islamic scholars believe that the Quran cannot be translated accurately, Muslims who are not native speakers of Arabic have to perform their daily prayers in Arabic even though they do not understand it. The fact that 1.4 billion Muslims listen to and recite the Quran on a daily basis extends the presence and influence of the Arabic language beyond its speakers in the Arab world. Moreover, because the Quran was revealed to the Arabs in their spoken language, many Islamic concepts stem from Arabic culture and these concepts are now shared by all Muslims. Hence, the Arabic culture is woven into the local cultures of all Muslims.

2.4 Linguistically

The internal structure of the Arabic language exhibits many interesting properties for linguistic theories. In section 2.4.1, we will discuss the nonconcatenative nature of Arabic morphology, the communicative function of subject pronouns, and the role that the structure of Arabic plays in syntactic constituency. In section 2.5.2 we discuss the unusual situation posed by the absence of native speakers of CA and MSA and in section 2.5.1 we discuss how Arabic discourse is affected by the oral tradition of the Arabic culture.

2.4.1 Implications of Arabic structure to Linguistic theories

The Nonconcatenative nature of Arabic Morphology Arabic, like other Semitic languages, is characterized by a complex and rich morphology. Traditional Arab grammarians recognized the root as the basic underlying form of Arabic words and described Arabic morphology الصَرفُ العرَبي *al-ṣarfu ʾl-ʿraby* in terms of patterns أوزَان *ewzān*. The building blocks of Arabic surface stems are the consonantal root which represent a semantic field like[KTB] 'writing' and a vocalism that represents a grammatical form. It is more often than not that the elements of the root and the vocalism when combined, form Arabic surface words that are discontinuous. For example the surface word "katab" 'wrote' can be represented as [KaTaB] where the upper case letters represent the elements of the root while the vocalism is represented by the lower case letters. The consonantal root means "writing" whereas the vocalism [a-a] has the grammatical meaning of the perfected verb. Unlike Romance and many other languages, Arabic morphemes are not contiguous. The building blocks of Arabic words are usually nested rather than concatenated, hence the nonconcatenative nature of Arabic morphology (McCarthy, 1981). The discontinuity of the root and the vocalism suggests that the well established definition of the morpheme as a meaningful minimal linguistic unit (i.e., one that does not contain a morpheme boundary within it) is not a comprehensive definition. This definition works well with languages with morphemes that are concatenated.

McCarthy proposes that Arabic words are formed by an association of the radicals (elements of the consonantal root) with the vocalism forming a prosodic template. Moreover, further work on Arabic morphology reveals its canonical-invariant property in Arabic templatic morphology (McCarthy and Prince, 1990). They show that Form 2 which is the causative of Form 1, is generated by reduplication of the second radical while keeping the canonical form invariant. Similarly, by changing the vocalism, the voice goes from active to passive while keeping the canonical form invariant. Inspired by these features of Arabic morphology, McCarthy and Prince developed a theory of prosodic morphology which defines templatic morphology in terms of the fundamental units of prosody such as the mora, the syllable, the foot and the phonological word. McCarthy also points out that the root in CA is subject to certain constraints. For example, there is a constraint that prohibits the reduplication of the first element in an Arabic root while allowing it for the second element.

Communicative Function of Subject Pronouns Arabic, like Italian, Spanish, Korean and many other languages allows for the deletion of subject pronouns when the information provided by the subject pronoun is recoverable. This led some linguists

(Perlmutter, 1972) to consider the subject pronoun in such languages as completely redundant, with no communicative function whatsoever, since it can be deleted with no loss of meaning. The optionality of the deletion of the subject pronoun in some languages led Chomsky to conclude in his "Theory of Principle and Parameters" (Chomsky, 1986) that the pro-drop parameter is part of universal grammar that is fixed by the early linguistic experience of a child. Languages with a pro-drop parameter may optionally delete subject pronouns when the condition of the recoverability of deletion is met (Chomsky, 1965). Other languages like English, French and German do not allow their subject pronouns to drop.

Elaborating on the insights of Eid (1980) subject pronouns in Egyptian Arabic are not redundant but have real communicative value. Farghaly (1982) shows that in Arabic sentences, the presence or absence of the subject pronoun could change the interpretation of a sentence. Consider the following two sentences:

(1) حَضَرَ علي وَانصَرَف هوَ *ḥaḍara ʿly wānṣaraf hwa* 'Ali came and he left'.

(2) حَضَرَ علي وَانصَرَف *ḥaḍara ʿly wānṣaraf* 'Ali came and left'.

The sentence in (1) is a compound sentence conjoined by the Arabic coordinator وَ *wa* meaning 'and'. Each of the two conjoined sentences consist of a verb and a subject. The subject in the first conjoined sentence is a lexical NP while in the second it a subject pronoun. However, the preferred reading of (1) is that the two subjects are disjoint in reference. The sentence in (2) is identical to the one in (1) except that the subject pronoun of the second verb is deleted. The deletion of the subject pronoun conforms to the recoverability of deletion condition because the features of the subject pronoun are present in the inflected verb. In contrast to (1), the preferred reading of the the sentence in (2) indicates that the subject of the two verbs in the sentence are coreferential. Thus, the presence of the subject pronoun in (1) signals the disjoint reference between 'Ali' and 'he' while the absence of the subject pronoun in (2) causes obligatory coreference. Farghaly (1982) proposed different structures for the two sentences above.

Eid (1980) argues convincingly that the subject pronouns in Egyptian Arabic could facilitate disambiguating ambiguous messages. Consider the following sentence:

(3) تحَدثَ الوزيرُ معَ الصحفي عندَمَا حَضَر *tḥadṯa 'l-wzyru mʿa 'l-ṣḥfy ʿndamā ḥaḍar*
'The minister talked to the journalist when he came'

The subject pronoun of the verb 'came' is deleted in this sentence. It could refer either to the journalist or to the minister. However, the preferred reading for (3) is that the pronoun refers to the journalist i.e. the most adjacent antecedent with which it agrees. In contrast, if the subject of the verb 'came' is explicit as in (4), the preferred reading would reference the furthest possible antecedent.

(4) تحَدثَ الوزيرُ معَ الصحفي عندَمَا حَضر هوَ *tḥadṯa 'l-wzyru mʿa 'l-ṣḥfy ʿndamā ḥaḍr hwa*
'The minister talked to the journalist when he came'

It would be interesting to see if the behavior of subject pronouns in Arabic is shared by all pro-drop languages and not restricted to Arabic. If other pro-drop languages do

not share these properties with Arabic, then we could gain a better understanding of the ways these languages vary.

Arabic Features and Syntactic Constituency Agreement in Arabic is very complex and encompasses a rich set of features including number, person, gender, humanness and definiteness. Moreover, Arabic exhibits anti-agreement; and agreement directionality plays an important role in defining certain syntactic categories. For example the first term of the Arabic noun construct which is equivalent to the English possessive phrase 'the cousin of the king' is almost always indefinite while the second term could be either definite or indefinite as shown below

(5) حُرَاسُ القَلعَةِ *ḥurāsu 'l-qalʿati*
guards the citadel
The guards of the citadel.

(6) *الحُرَاس القَلعة *al-ḥurās al-qalʿt*
the guards the citadel

The only difference between the grammatical noun construct in (5) and the ungrammatical phrase in (6) is that in (6) the first term of the construct is definite while in (5) it is indefinite. The so called "equational sentences" in Arabic consist of a subject and a predicate. The subject is a noun phrase while the predicate could be an adjectival or a noun phrase. The subject has to be definite while the predicate must be indefinite. If the definiteness relationship between the subject and the predicate is reversed, the resulting constituent is no longer a sentence. It becomes a noun construct. Consider:

(7) الرَجُلُ قَاتِلُ *al-rağulu qātilu*
the man killer
The-man is a killer.

(8) قَاتِلُ الرَجُلِ *qātilu 'l-rağuli*
killer the-man
The killer of the man.

The following sentence shows that in Arabic quantifiers have to disagree in gender with the noun they quantify.

(9) اشتَرَيتُ بضع آزهَار *āštaraytu bḍʿ ʾāzhār*
bought-I some flowers
I bought some flowers.

The quantifier بضع *bḍʿ* here is masculine and the noun it quantifies is feminine because Arabic does not allow the quantifier to agree in gender with the noun it quantifies.

Subject-verb agreement in SVO sentences is different from subject-verb agreement in VSO sentences (Fehri, 1993). For example: when the subject precedes the verb, the verb agrees in gender, person and number with the subject. But when the verb precedes the subject the verb has to be singular regardless of the number of the subject. The agreement system of Arabic raises interesting questions about agreement triggers, anti-agreement and the role of the directionality of agreement.

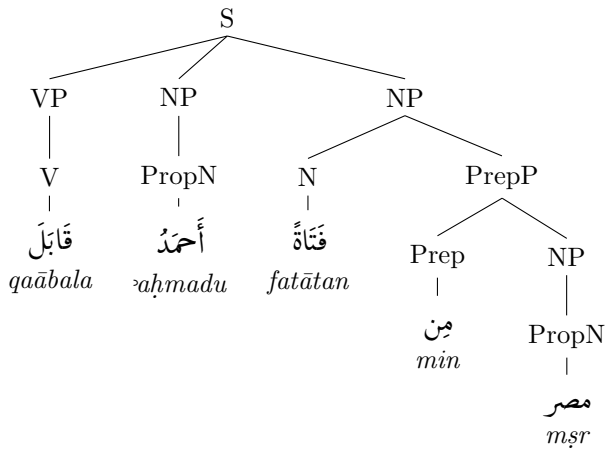

FIGURE 1 The syntactic structure of senetnce (10)

2.5 Prepositional Attachment

Prepositional phrases in Arabic may attach to a verb or to a noun phrase. This creates the problem of identifying case when the prepositional phrase modifies the verb or when it modifies the noun. Consider the following two sentences where the prepositional attachment problem causes confusion in the English equivalent.

(10) قَابَلَ أَحمَدُ فَتَاةً مِن مِصر *qābala ʾaḥmadu fatātan min miṣr*
met Ahmed girl from Egypt
Ahmed met a girl from Egypt.

(11) قَابَلَ أَحمَدُ فَتَاةً فِي مِصر *qābala ʾaḥmadu fatātan fiy miṣr*
met Ahmed girl in Egypt
Ahmed met a girl in Egypt.

In sentence (10) the prepositional phrase مِن مِصر *min miṣr* 'from Egypt' modifies the noun. We can substitute the phrase فَتَاةً مِن مِصر *fatātan min miṣr* 'a girl from Egypt' with the phrase فَتَاةً مِصرِيَةً *fatātan miṣriyatan* 'an Egyptian girl'. In contrast, we cannot do the same in sentence (11) because the meaning in (11) does not imply that the girl that Ahmed met in Egypt, was an Egyptian. She could be American, Lebanese or from any country including, but not limited to Egypt. The correct analysis would attach the prepositional phrase in (10) to the noun while the prepositional phrase in (11) would be attached to the verb phrase. The difference in the syntatcic structures of (10) and (11) can be seen in Figures 1 and 2 respectively.

2.5.1 Repetition and Redundancy in Arabic Discourse

An understanding of the nature and properties of Arabic discourse would facilitate machine translation systems when dealing with a written text or transcribed speech since Arabic discourse and modes of argumentation are very different from Western norms (Johnstone, 1991, 1983). Johnstone (1991) examined several expository texts in contemporary Arabic. She noted that the texts written by highly educated Arabs, were

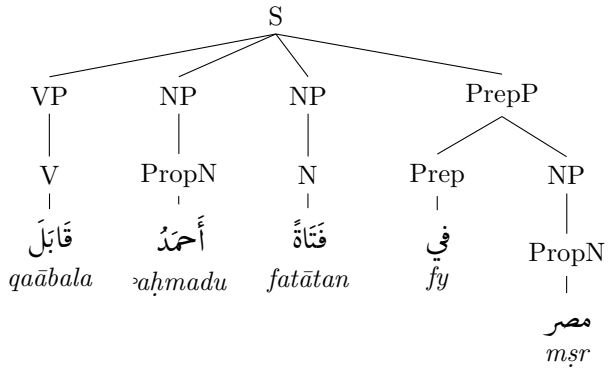

FIGURE 2 The syntactic structure of senetnce (11)

characterized by elaborate patterns of lexical, morphological and syntactic repetitions. She also confirmed from her personal experience that "Arabs convince people: namely by repeating" (Johnstone, 1991). She noted that the texts she examined were highly repetitious. For example, there was frequent use of paraphrase, parallelism and lexical couplets, usually conjoined, two-word phrases, that are completely or nearly synonymous. Examples of these couplets are:

- التَّأيِيدُ وَ المُسَاندةُ *al-taʾāyyydu wa 'l-musāndtu*
  aid and assistance
- الوَهمُ وَ الخَيَالُ *al-wahmu wa 'l-ḫayālu*
  illusion and imagination
- التخريبُ وَ التَّدميرُ *al-tḫrybu wa 'l-tadmyru*
  destruction and demolition

Johnstone concludes that Arabs, unlike Westerners, persuade others through repetition and not through logical reasoning. She investigated the origin of this style of persuasion. She cites other studies (Bauman, 1977, Stankiewicz, 1960) that associate repetition and the focus on form rather than content, with poetry and oral discourse. However, what is perceived as "meaningless repetition" by Westerners, is highly regarded by Arabic speakers. For them, it is an embodiment of the beauty and elegance of Classical Arabic. The coherence, rhythm and rhetoric contribute to the effectiveness of the argument. But Arabs do not assume that the elegance of an expression alone is proof of its validity as has been claimed by Bateson (Bateson, 1967). The elaborate repetitions in contemporary written Arabic discourse can be explained by the oratorical nature of the Arabic culture as we discussed in section 2.3.1. A comparison was made between discourse in literate societies and discourse in oral cultures. It was found that literate cultures rely on the factual accuracy of a message and favor evidence, reasoning, and analysis. All information is included in the message and the burden is on the speaker to provide the proof for his argument. In contrast, oral cultures rely on emotional resonance. They place higher emphasis on symbolism and they are less rational, adhering to a more intuitive approach. For example, a single anecdote can constitute

evidence for a conclusion. The burden is on the listener to complete and decipher the coded message.

2.5.2 The Absence of the Native Speaker

There is has been considerable debate around the question of whether Classical Arabic is a natural language or not and whether it is worthy of investigation within the generative framework (Elgibali, 1996). Chomsky made it clear that linguistic theory is only concerned with characterizing native speakers intuitions about their language (Chomsky, 1965). He says, "Linguistics theory is concerned primarily with an ideal speaker-hearer, in a completely homogeneous speech community, who knows its language perfectly". This statement eliminates Classical Arabic and MSA from the scope of generative grammar because it has often been stated that Classical Arabic is not the native language of any present group of people. Arabic speakers learn Classical Arabic at school in the same way they learn any other foreign language. Because Classical Arabic is not acquired naturally, speakers of Arabic do not have an inherent intuition of Arabic grammar. Chomsky (1965) postulated that the goal of linguistic theory is to describe and define the linguistic knowledge of native speakers. To adhere to this principle, linguists rely heavily on the grammatical judgments that they elicit from native speakers.

In the absence of native speakers, as is the case with Classical Arabic and Modern Standard Arabic, linguists who study these versions can neither test nor validate their theoretical assumptions. As a result, many linguists have shifted their attention to the colloquial versions of Arabic where they have access to native speaker's intuitions about their language.

However, a natural language may not be defined so narrowly: that is, as a language that has living native speakers. Some endangered languages lose the last native speaker of the language. This fact does not transform the language from natural to unnatural. Latin is a natural language although it is not spoken by any present day group of people nor is it acquired naturally because people learn it as a second language.

Artificial languages such as Esperanto and programming languages are created at a particular time by identifiable authors but are neither linked to a culture nor a literary tradition. In contrast, neither Classical Arabic nor Modern Standard Arabic are associated with a definite point of origin nor are they attributed to any particular author. They arose naturally to fulfill the needs of their speech community and as importantly, are tied to a specific culture and an impressive literary and knowledge-seeking tradition. It is not surprising then, that Classical Arabic has survived through the centuries, while the artificial language, Esperanto, has not.

3 Arabic Linguistics

The only complete grammar of Arabic to date, is that written by early Arab grammarians. This grammar, written more than a thousand years ago still serves as the main reference even when used to describe Modern Standard Arabic. Most current work in Arabic computational Linguistics relies at least to some degree on the analyses made by those pioneering Arab grammarians (see chapter 4 in this volume).

In this section we begin by presenting the assumptions and goals of the Arabic linguistic tradition. Then we discuss their methodology while comparing it with the methodology of modern linguistics theory in section 3.2. We present some of their descriptions of CA in section 3.3 and show that while their description of Arabic was pioneering and very advanced at the time it was developed, it falls short of the requirements for computer processing.

3.1 Goals and Assumptions of the Arabic Linguistics School

The Arabs did not seriously study and analyze their language until they converted to Islam and felt they were privileged and honored to have the Quran revealed in their language by the Prophet Mohammed. The decision to initiate research in Arabic grammar was made at the highest level of government by the fourth Caliph Ali Ibn Abi Talib علي ابن ابي طالب *ʿly ābn āby ṭālb* who ruled over the Islamic Caliphate from 656 to 661 CE. Versteegh (1997b) cites a very interesting story of how the Caliph was concerned about the purity of the Arabic language and wanted to set norms for its usage. The story that was given by Abu l-Aswad Al-Du'ali, an early Arab grammarian, is as follows:

> I came to the commander of the Believers أَميرُ المُؤمنين *ʾamyru ʾl-muʾmnyn* 'Ali Ibn Abi Talib' - may God have mercy on him - and saw in his hand a manuscript. I said to him "What is this, Commander of the Believers?" He said: "I was reflecting on the language of the Arabs and noted that it had been corrupted by our mixing with these red persons - i.e. foreigners - and I wanted to make something for them on which they could fall back on which they could rely". then he handed me the manuscript, and I saw that it said: "Language is a noun and verb and particle. The noun is what informs about a named object; the verb is that with which the information is given; and the particle is what comes for a meaning. He said to me: Follow this direction and add to it what you find (Versteegh, 1997b).

3.1.1 Goals of Traditional Arabic Grammarians

The linguistic analysis of Arabic by early Arab grammarians contains clear objectives. Their objectives were:

1. Proper understanding and interpretation of the Quran and the Prophet's sayings الأَحَاديثُ النَبَويَةُ الشَريفَةُ *al-ʾaḥādyṯu ʾl-nabawyatu ʾl-šaryfatu*. The Quran contains instructions that closely relate to Muslims' daily life: instructions that regulate marriage, inheritance, divorce, second marriage of women, polygamy, Zakat, etc., are defined in the Quran. However, common followers of Islam have always needed help from specialists on how to observe Jurisprudence in their daily lives. These specialists needed a grammar of the language to help them interpret the Quran. As a result, the primary objectives of the Arab grammarian were to describe and explain the grammatical structure of Arabic.
2. Provide Arabic speakers with phonetic and phonological descriptions to help them recite the Quran correctly. They studied the sound system of Arabic consonants and vowels describing their place of articulation and manner of articulation while accounting for the phonological processes of assimilation, elision, vowel shortening,

pharyngealization, stress and intonation, etc.

3. Protect the Arabic language exemplified by the Quran, pre-Islamic poetry and the spoken language of the Arab Bedouins from corruption, by providing norms for correct usage. They designed grammars and lexicons of Arabic that define well formednesss and distinguish the correct usage from the faulty Arabic spoken by the new converts to Islam. Ibn Khaldoun, an Arabic historian, points out "Grammar was necessary when the new Muslims threatened to corrupt the Arabic language by their mistakes" (Ibn Khaldun, 1958).

The objectives of early Arab grammarians were thus, more applied. Their close adherence to the Quran as a corpus made them begin with what is practical before establishing theoretical foundations for each area of research. (El-Raghi, 1979).

3.1.2 Assumptions of Early Arab Grammarians

The intellectual, political and religious thinking of the Islamic Caliphate had a profound effect on the way Arab grammarians regarded the language they were describing. Some of the assumptions they made are as follows:

Supremacy of the Arabic Language Arab grammarians believed that the Arabic language was superior to any other language in the world. They based their belief on two grounds: one religious and the other linguistic. The religious factor, is that Arabic was the language of God because the Quran was revealed to the Prophet in Arabic. Thus, the Arabic language is sacred and they believed that every Muslim should learn the Arabic language. Ibn Faris, a great Arab grammarian says (translation is mine) "I say that learning the language of the Arabs is a must for anyone who is interested in the Quran or the Sunna. This knowledge of the Arabic language is indispensable. This is because the Quran is revealed in the language of the Arabs; and the Prophet, peace be upon him, is an Arab. Thus for anyone who wants to learn what is in the holy book and what is in the Sunna of his Prophet, peace be upon him, will find no way except by learning the Arabic language". Ibn Faris states further, that the Arabic language is superior to other languages because it exhibits frequent lexical equivalences, metaphor, assimilation, preposing and postposing, etc. Arab grammarians thought these linguistic properties were unique to the Arabic language. In contrast, modern linguistic theory is based on the assumption that no language is better than another and there are many features that are common to all languages. Even when languages vary, they vary within limits. The focus in modern linguistics then, is on characterizing language universals.

Origin of Language Arab grammarians debated the question of whether language was inspired by God or whether it arose through mutual agreement and convention. According to the Quran "God taught Adam all names". This could also be interpreted to mean that God gave Adam the ability of naming i.e. the language faculty. This controversial issue was discussed at length by the famous Arab grammarian Ibn Jinni in his famous book اَلخَصَائِص *al-ḫaṣāʾiṣ* 'The Special Features' written at the end of 10th century AD.

Variability of Surface Structure Arab grammarians working on the fixed corpus of the Quran, the pre-Islamic poetry and the trusted, uncontaminated intuitions of the Bedouins posited rules deduced from that corpus. They described Arabic verbal

sentences as having the order Verb, Subject, Object. However, they also noted that other orders existed such as Subject Verb Object or even Object Verb Subject and Verb Object Subject. To account for this variability, they claimed that any order other than VSO is derived from an underlying order أَصل *ʾaṣl* through preposing تَقديم *taqdym* and postposing تأَخير *tʾaḫyr*. Thus there is a resemblance between the Traditional Arabic school and Transformational Generative (TG) grammar which describes the structure of language at two levels: surface structure and deep structure.

Lack of Interest in Other Languages Arab grammarians did not care to look seriously at other languages or compare them to Arabic. Had they done so, they would have discovered, that Hebrew and Arabic, for example, are very close morphologically and syntactically.

Lack of Interest in the Dialects Like all traditional grammars, Arab grammarians thought that speech was inferior to writing and that the literary or poetic language was the "best" language. Thus, for many centuries Arab grammarians neglected the study of Arabic dialects. In fact, their objective was to "protect" CA from the errors made by the learners of Arabic. They reported on these "errors" that probably resulted from interference from the mother tongue of the new Muslim coverts when they tried to speak Arabic. Had they documented the language change that was happening during this time, we would have a better idea of how Arabic dialects arose in the first place.

No Diachronic Study of Language Arab grammarians did not consider language as an organism that changes over time. Therefore, they did not make any comparisons of the Arabic language at two different periods so no diachronic study of Arabic was performed.

3.2 The Methodology of Traditional Arabic Linguistics

Corpus-based Arab grammarians focused on a fixed corpus to discern the the structural regularities underlying this corpus. Their corpus consisted of the Quran, pre-Islamic poetry and the sayings of the Prophet. So their approach was descriptive rather than prescriptive in spite of their aim to set norms for correct usage in Arabic. They never imposed their views of the correct usage without having evidence from the corpus. However, they often resorted to consultation with native Arabic speakers to elicit their intuitions about the grammaticality of a sentence or a phrase. They defined native speakers as those Arab Bedouins who did not come in contact with the sedentary population. Thus, their language was still pure and uncontaminated. For example, Sibawayhi, the great Arab grammarian and author of the first comprehensive treatment of Arabic grammar frequently consulted Arab Bedouins for their intuitions. This practice of consulting the Arab Bedouins was used by Arabic grammarians and was limited to the early period of the Islamic Caliphate.

Governance The notion of Governance plays a central role in the syntactic description of Arabic. Case assignment is described and explained by Arabic grammarians through the notion of governance. All nouns and adjectives are governed in Arabic. Arabic nouns

and adjectives, with the exception of diptotes المَمنوعُ مِنَ الصَرفِ *al-mamnwʿu mina 'l-ṣarfi*, are assigned one of three cases depending on the function the noun or adjective in the sentence. The three case endings for nouns and adjectives are:

1. A high round vowel ضَمَة *ḍamat* corresponds to Nominative referred to in Arabic grammar as اَلرَفع *al-rafʿ*
2. A low from vowel فَتحَة *fatḥat* corresponds to Accusative referred to as اَلنَصب *al-naṣb*.
3. A high from vowel كَسرَة *kasrat* corresponds to Genitive known as اَلجَرّ *al-ǧarr*

In principle the three main word categories: nouns, verbs or particles could be governors and therefore may assign case. A governor may govern multiple governees while a governee may have one and only one governor (Versteegh, 1997b). For example a verb governs its subject(s) and assigns it the nominative case and also governs its direct object and assigns it the accusative case. Nouns and adjectives could be governors and governees at the same time. Consider:

(12) شَاهَدتُ مُديرَ البَنكِ *šāhadtu mudyra 'l-banki*
saw-I manager the-bank
I saw the manager of the bank

The verb 'saw' in the above sentence assigns accusative case to 'manager'. Thus manager is a governee here. It is also a governor since it is the first term of the noun construct 'manager of the bank' and assigns genitive case to the second term of this noun construct المُضَاف اَلَيهِ *al-muḍāf ālayhi*.

Proper understanding of the Quran and Arabic poetry relies to a large extent on identifying the correct case especially for the arguments of predicates because of the relatively free word order of Arabic and the frequent preposing and postposing of constituents.

Surface and Deep Structure Arab grammarians noticed that in some texts they could discern meanings that were not explicitly mentioned in the text. For example in the following sentence:

(13) ضَرَبَ زَيدًا *ḍaraba zaydan*
hit-past Zaydan
He hit Zaid.

They defined it as a verbal sentence because it begins with a verb and is followed by a direct object that is governed by the verb and is therefore assigned the accusative case. This sentence does not have an explicit subject so is a sentence whose subject pronoun is dropped. However, the Arabs did not know about the pro-drop parameter. Their analysis was that there was a subject, but the subject was hidden مُستَتِر *mustatir* and that it is interpreted as 'he' because of the declension of the verb. They followed this reasoning when a constituent appeared at position different from the expected grammatical position. It is moved from its position in the "deep" structure to the

"surface" structure. By positing an underlying level of analysis, they were able to achieve more generality for their syntactic rules.

3.3 The Description of Arabic by Traditional Arab Grammarians

In this section we will focus on the classification of the different types of Arabic sentences and how the correct definition of phrasal constituencies, neglected in traditional Arabic grammar, are crucial in Arabic Computational Linguistics.

3.3.1 Sentence Types

Identifying basic constituents of a sentence is important in machine translation, information extraction and retrieval, event and relation extraction, question answering systems, and data mining. We define basic constituents as the phrasal building blocks of a sentence such as subject, object, verb, adverbials and any other constituents that help define who did what to whom, when, where and for what reasons. Thus, analyzing a sentence to describe such constituents is important for computational programs whether based on linguistic engineering or machine learning technology. Correct classification of sentence type in a language, is an essential step in that direction.

Arabic grammarians divided Arabic sentences into two types.

Verbal Sentences They defined verbal sentences as sentences that begin with a verb such as:

(14) هُزِمَ الأَعدَاءُ *huzima 'l-ʾaʿdāʾu*
defeated=passive the-enemies
The enemies were defeated.

(15) أَعلَنَ الوَزيرُ أَنَّهُ سيحضُرُ الحفلَ *ʾaʿlana 'l-wazyru ʾanahu syḥḍuru 'l-ḥfla*
announced the-minister that-he will attend the party
The minister announced that he will attend the party.

Sentence (14) is a verbal sentence as it begins with a verb. The sentence is passive and the grammatical subject is not the agent but is the grammatical object of the corresponding active sentence. Sentence (15) is interesting because though it is a verbal sentence beginning with a verb and an overt subject, it has a sentential complement with an explicit complementizer أَنَّ *ʾanna* 'that'. The sentence that follows the complementizer is not a verbal sentence because it does not begin with a verb although it has a verb, a subject and a direct object.

Nominal Sentences A nominal sentence is a sentence that begins with a noun such as:

(16) الحربُ مُدَمِرةٌ *al-ḥrbu mudamirtun*
the-war destructive
The war is destructive.

(17) الفيدرَالية عقَدَت مؤُتمرًا صحفيِيًا أَمس *al-fydrālyt ʿqadat mʾutmran ṣḥfyiyan ʾams*
the-federation held conference press yesterday
The federation held a press conference yesterday.

Both (16) and (17) are the same type of sentences since both begin with a noun. Yet, they are very different. (16) is usually analyzed as a subject followed by a predicate. The predicate could be a noun phrase, adjectival phrase, or a prepositional phrase. It does not have an overt verb and its time refers to the present time. (17) on the other hand is described as beginning with a topic, followed by a verbal sentence which in turn is analyzed as a verb, with a hidden subject followed by a direct object and a time adverbial. The reader of the sentence has to make the connection between the hidden subject of the verbal sentence and the topic of the nominal sentence. While identifying the main constituents of a sentence might be possible within the Classical Arabic framework because Classical Arabic texts usually have overt case markers, this is not possible in Modern Standard Arabic texts. MSA texts are characterized by the absence of the diacritics that mark case. In spoken MSA, speakers tend to pause at word endings to save themselves embarrassment if they confuse case endings. MSA computational linguistics applications would benefit greatly from a description based on more formal and explicit pointers rather than on cases that do not appear in the surface structure or have hidden subjects that need to be recognized. In the following section we will show that for computational and linguistic considerations, a different classification and description of Arabic sentences would be more computationally efficient.

3.3.2 Phrasal Constituents in Traditional Arabic Grammar

Describing and explaining the different endings of an Arabic word had the highest priority for Arabic grammarians. Reciting the Quran correctly and interpreting its meaning depends crucially on knowledge of the rules of parsing قَوَاعِدُ الإِعرَابِ *qawāʿidu 'l-ʿirābi* which are described in terms of governance as shown above. Identifying phrasal categories below the sentence level such as noun phrases, adjectival phrases and their internal structure did not receive as much attention as case ending اَلإِعرَاب *al-ʿirāb* received.

Recognizing Arabic phrasal categories and their syntactic properties and boundaries is extremely important and proved to be a difficult problem in Arabic computational linguistics. Below is an example of a very common Arabic structure known as اَلإِضَافة *al-ʿiḍāft*; sometimes referred to as the noun construct. Consider the following:

(18) مَقَرُ الوِزَارةِ *maqaru 'l-wizārti*
headquarter ministry
The headquarters of the Ministry.

(19) جَميلةُ الوَجهِ *ǧamyltu 'l-waǧhi*
pretty-fem thh-face
with the pretty face.

(20) أَمَامَ المنزِلِ *ʿamāma 'l-manzili*
front the-house
In front of the house.

Arabic traditional grammarians noted that in all the above sentences, the second term is governed by the first and is assigned a genitive case so they grouped them together as an idaafa. In an Arabic machine translation system, we must be able to distinguish these three phrases by one of the following analyses: (18) is a noun phrase,

(19) is an adjectival phrase and (20) is a prepositional phrase. Such an analysis is more relevant and useful in ANLP applications than for the traditional analysis which treats them as one structure based on case ending which is not applicable for texts in MSA. Arabic computational linguistics would benefit greatly from surface-based grammars of MSA the same way NLP systems benefited from lexical-based grammar formalisms such as Lexical Functional Grammar (Bresnan, 2000) and Head Phrase Structure Grammar (Pollard and Sag, 1994).

4 Arabic Computational Linguistics

Arabic computational linguistics is part and parcel of mainstream computational linguistics. As such the prevailing paradigms in Arabic computational linguistics are the same as those found in the field, in general. This volume contains reports on work in both paradigms and on the hybrid approach. Thus we begin this section with a brief description of the two paradigms. In the following section we describe and compare the goals of developers of Arabic NLP systems in the Arabic speaking world and the goals of developers in a non-Arabic environment. In section 4.3 we briefly comment on computational treatments of Arabic morphology while in section 4.5 we review Arabic sentence analysis of the Penn Arabic Treebank.

4.1 The Prevailing Paradigms in the Field

Computational linguistics, like any other discipline, exhibits cyclic change with respect to prevailing paradigms (Oepen et al., 2000). In the late 1970's and 1980's new technologies emerged to make linguistic engineering more efficient. The new technologies targeted two areas: processing efficiency and linguistic descriptions. In the area of computer processing more efficient techniques were developed such as chart parsing (Kay, 1973), definite clause grammars (Pereira and Warren, 1980), and unification (Shieber, 1986). At the linguistic level, monostrata formalisms such as head phrase structure grammar (HPSG) (Pollard and Sag, 1994) proved to be very useful in designing grammars with very wide coverage. However, since 1990's we have seen a dramatic shift in the field from linguistic engineering to statistical approaches (Manning and Schuetze, 1999, Jurafsky and Martin, 2000, Baldi and Brunak, 2001, Joachims, 2002, Abe, 2005, Bishop, 2007, Cherkassky and Mulier, 2007). There were several factors that paved the way for this change to take place.

1. It is impossible to encode common sense knowledge, knowledge of the world and the context that native speakers use to communicate this knowledge. In the absence of such knowledge, NLP systems have to consider a huge number of logically possible analysis that never arise in the minds of native speakers. Considering all these logical possibilities slows down processing of input and systems within the symbolic framework are usually slow.
2. Grammar engineering is expensive especially when going beyond the stage of developing toy grammars. Grammars for coverage of sizable data fields suffers greatly from rule interaction which slows processing.
3. Many computational linguists were more concerned with strict adherence to linguistic theories than developing systems with practical applications. Instead,hey

were obsessed with theoretical correctness.

4. The web became the primary source of data needing to be processed. Blogs, emails, chats and web pages are full of errors, typos, incomplete sentences posing a challenge for parsing.
5. Rule-based machine translation systems suffered from lack of fluency in the target language and translated texts suffered from ungrammatical phrases and sentences.

The development of successful speech recognition systems using probability theory and machine learning approaches paved the way for a new empirical approach (Manning and Schuetze, 1999) while proving its cost effectiveness, validity and robustness. As a result, statistical approaches to natural language processing have dominated the field since the 1990's. The new paradigm is characterized by the following features:

1. The use of training and testing data. The closer the testing data to the training set, the better the results.
2. The use of both supervised and unsupervised learning methods.
3. Based on probability theory it looks at language and cognition as probabilistic and predicts the next word given the previous one.
4. It assumes that linguistic knowledge is present in linguistic data and that machine learning techniques can extract this knowledge through cycles of training and retraining until it "learns" the language.
5. Development time is short compared to rule-based systems.
6. The statistical models are robust.
7. The use of language modeling which gives the output the fluency that is absent from symbolic systems.
8. The use of bilingual parallel corpora and translation memory in machine translation.
9. The use of n-gram models
10. The development of algorithms that are language independent.

However, there are problems with statistical approaches to NLP. First, performance deteriorates rapidly when input varies from the training data. In contrast, the performance of symbolic systems is usually consistent. Second, in some cases it is difficult to predict the kind of data a system will receive. Thus, training such a system is difficult. Third, the language of emails, chat, and blogs are noisy, affecting the system's performance. Fourth, there could be a point where adding more training data would confuse the system; known known as the threshold problem. Fifth, when a phenomenon occurs sparsely there is insufficient data to "learn" from. This is known as the "sparse data problem". On another level, prominent computational linguists (Zaenen, 2006, Reiter, 2007, Jones, 2007) have recently expressed misgivings regarding the exclusive reliance on machine learning approaches while neglecting the contributions of symbolic language processing. Thus, research in Arabic computational linguistics has been influenced by the same philosophical shift as the paradigms in natural language processing. Bender (2009) argues that even when developing language-independent NLP systems, generalizations from the typology of language which defines the ways languages vary from each other is very much needed.

There is a growing realization that incorporating linguistic knowledge in statistical NLP systems improves its performance. Please see chapters 7, 9 and 11 in this volume.

4.2 Goals of Arabic Computational Linguistics

Arabic computational linguists developing systems for Arabic speaking users usually have a different focus than those developing Arabic NLP software enabling non-Arabic speaking people to understand Arabic texts and speech.

Objectives of Arabic NLP Applications in a Non-Arabic Environment

1. To develop tools that enable English speakers with limited or no knowledge of Arabic deduce the gist or the general meaning of Arabic texts.
2. To be able to extract pieces of information of interest such as person names, addresses, phone numbers, email addresses, etc., from Arabic texts.
3. Produce tools that can comprehend a limited set of spoken Arabic phrases and short sentences and that are capable of producing spoken short Arabic utterances on demand.
4. Develop high quality Arabic to English machine translation systems.

Clearly, the emphasis is on developing Arabic language capabilities for non-Arabic speaking individuals. We will see below that the objectives of Arabic computational linguistics in the Arabic speaking world is quite different.

Goals of Arabic NLP Applications in the Arabic Speaking World

1. Transfer of knowledge and technology to the Arab World. Most recent publications in science and technology are published in the English language and are not accessible to Arab readers who have no competence in English. For humans to translate this huge amount of material to Arabic is very costly and time consuming. Arabic NLP could reduce the time and cost of translating, summarizing and retrieving information in Arabic for Arab speakers.
2. Modernize and fertilize the Arabic language. This follows from (1) above. Translating new concepts and terminology into Arabic involves coinage, arabizations and making use of lexical gaps in the Arabic language. This will have a dramatic effect on the revitalization of the Arabic language, allowing it to fulfill essential needs for its speakers.
3. Improve and modernize Arabic linguistics. Arabic NLP needs a more formal and precise grammar of Arabic than the current traditional grammar. Innovation is needed while preserving the valuable heritage of traditional Arab grammarians.
4. Make information retrieval, extraction, summarization and translation available to the Arab user. The hope is to bridge the gap between peoples of the Arab world and their peers in the advanced countries. By making information available to the Arabs in their native language, Arabic NLP tools empower the current generation of educated Arabs. Thus Arabic NLP tools are indispensable in the struggle of Arabs to keep pace with the rest of the world. This is a matter of national security for the Arab World (Farghaly, December 2008).

4.3 Computational Arabic Morphology

Arabic morphology has received much attention from engineers and computational linguists since the late 1970's and early eighties. Pioneering work on the computational morphology of Arabic focused on the retrieval of the consonantal root from fully inflected words (Hlal, 1979, 1985) through the process of decomposition and repeated matching against a dictionary of roots. Arabic morphology poses a challenge from a theoretical point of view (McCarthy, 1981, Farghaly, 1987, 1994). For example, McCarthy (1981) proposes a two tier notation where the elements of the consonantal root are on one tier and the vocalism is on a separate tier. Farghaly (1987) says that Arabic surface words must be analyzed at three levels: the level of the unpronounceable root, the level of the vocalism and the level of the affixes. Farghaly (1994) suggests that the Arabic lexicon may consist of underspecified entries to represent the discontinuous nature of Arabic morphemes. The question of what constitutes entries in the Arabic lexicon has not been resolved yet. Most Arabic morphological analyzers including the most advanced treatment of Arabic morphology at Xerox Research Center in France embrace the view that the root is the basic entry in the Arabic lexicon (Beesley, 2001). Recently, it has been shown that a stem-based approach to Arabic morphology would be more relevant for Arabic computational linguistics applications (Buckwalter, 2002, Farghaly and Senellart, 2003). Positing the stem as the main entry in the Arabic lexicon eliminates the step of generating stems from roots. Stems are then associated with appropriate morphological, syntactic and semantic features that are needed in the syntactic analysis of Arabic. But most current morphological systems do not go beyond the strict morphological analysis of Arabic words. The morphology syntax interface is often neglected as has been noted in chapter 5 in this volume. For example, most morphological analyzers of Arabic do not categorize auxiliary verbs in Arabic as such. A verb like يَجِب *yağib* 'must' and many others for example, do not inflect for the first person nor for the feminine singular /* أَجِب *ʾağib* /* تَجِب *tağib*. Such information is crucial for checking agreement to identify syntactic categories. There are also verbs that serve more than one function. For example, the auxiliary verb كَانَ *kāna* functions as past tense marker in nominal sentences such as كَانَ أَحمَدُ في مِصر *kāna ʾaḥmadu fy miṣr* 'Ahmed was in Egypt', while it functions as a progressive aspect marker when it is followed by a verb as in

كَانَت تُغَني عندَمَا رَأَيتُهَا *kānat tuġany ʿndamā raʾaytuhā*
'She was singing when I saw her'.

Information about stems that is relevant for syntactic analysis should be included in any adequate morphological analysis of Arabic. It is not sufficient to correctly isolate stems from all affixes. What is needed is to highlight morphological information that can be relevant in the syntactic and semantic analysis of Arabic.

4.4 Morphology and Arabic Speech Recognition

Evidence from Arabic morphology seems to suggest that in the majority of Arabic words, Arabic consonants have an exclusive semantic function that differs from that of the vowels. Consonants in an Arabic stem almost always alone carry the lexical meaning of that stem. Vowels, on the other hand, have the function of making grammatical

distinctions. Consider the following list:

1. qatala 'he killed'
2. qatiil 'killed singular'
3. qaatil 'killer'
4. muqaatil 'fighter'
5. qattala 'he over killed'
6. qitaal 'fighting noun'
7. qitaali 'fighting adj'
8. qatl 'the act of killing'
9. maqtal 'fatal wound' or 'the killing of'
10. qtlaa 'killed plural'

The above is but a subset of possible stems that can be generated from the root [QTL] meaning 'killing'. Derivations of different forms do not alter the members of the root nor their order. Instead, derivations are accomplished by modifying the vocalism. Each stem has a different vocalism while the consonants of the root remain constant. This well known and attested observation has profound implications for Arabic speech recognition and Arabic language teaching. Arabic speakers seem to have intuitions about the role of consonants and vowels in their language. Unlike speakers of other languages, Arabic speakers can afford to pay less attention to the vowels than the consonants since only consonants carry the semantic meaning which is more crucial to understanding the message than is the grammatical information carried by the vocalism. The Arabic word for 'here' is /huna/ in MSA, /hina/ in Egyptian Arabic, /hoon/ in Levantine Arabic and /hni/ in Gulf Arabic. An Arabic speaker hearing this word by a speaker of another dialect does not need to think twice before relating what he heard to the equivalent in his dialect because they share the same consonants and differ only in the vocalism. The implications for speech recognition is that training Arabic speech recognition systems would be different from training the system for other languages. Training for Arabic could assign more weight to consonants than to vowels.

4.5 The Penn Arabic Treebank

In this section we will comment briefly on the most important resources that have become available to researchers in Arabic computational linguistics. In the last ten years, there has been excellent and much needed work at the Linguistic Data Consortium of UPenn for compiling and annotating various corpora of Modern Standard Arabic and the dialects. These resources have become available to researchers and developers in academia and the industry. The following are a few comments on the latest Arabic Treebank guidelines released in August 2009 (Maamouri et al., 2009).

4.5.1 MSA Sentence Types

The Arabic Treebank (ATB) guidelines provide an excellent definition of the nominal sentence in MSA. The nominal sentence is one that consists of a subject and a predicate but lacks a verb. This definition is different from the traditional Arab grammarians' definition of the nominal sentence: one that begins with a noun whether it contains a verb or not. The ATB definition eliminates, correctly, the traditional definition that

classifies each of the following sentences as a different type although syntactically they are similar.

(21) اوبَامَا زَارَ القَاهِرَة *āwbāmā zāra 'l-qāhirat*
Obama visited Cairo
'Obama visited Cairo '.

(22) زَارَ اوبَامَا القَاهِرَة *zāra āwbāmā 'l-qāhirat*
visited Obama Cairo
'Obama visited Cairo'.

Both (21) and (22) have a verb, a subject and an object. However, traditional Arabic grammar considers (21) to be a nominal sentence because it begins with a noun while it regards (22) to be a verbal sentence because it begins with a verb. The ATB guidelines, correctly, regards both sentences to be verbal because both contain a verb.

However, one would expect that the ATB guidelines would move to the logical conclusion and regard "Obama" in both sentences as being the subject. Unfortunately, this is not the case. Under the ATB guidelines, "Obama" would be the subject in (22) but only the TOPIC in (21). Then a trace would follow the verb with an index pointing to the TOP which is another way of saying that what has been categorized as TOP is actually the subject. There is no convincing linguistic evidence that would prevent us from categorizing "Obama" in (21) as the subject of the sentence. Actually only subjects that precede the verb would have full agreement with the verb. There is no reason to impose an imaginary deep structure order on all surface realizations. It would be more useful for Arabic computational linguistics applications to identify the subject and objects of Arabic sentences regardless of where they occur in the surface structure. Whether the subject or object occurs before or after the verb does not change its function in the sentences. It is this function that is crucial for understanding who did what to whom.

4.5.2 Pro-drop Sentences

When the subject or direct object appears before the verb in an Arabic sentence, the ATB posits a trace inside the VP following the verb with the same index of the "preposed" subject or object. The idea, of course, is to show that the "original" position of the subject is inside the VP following the verb. The trace is an indication that the subject was in the original place and then moved to the front. Such analysis is somewhat consistent with the analysis in traditional Arabic grammar which considers the subject in these sentences as "hidden" ضَمِير مُستَتِر *ḍamyr mustatir*.

There is another case when the subject is completely dropped from the sentence as the example given on page (13) in the guideline which we present below in (23). This sentence is a subjectless sentence. Since there is no subject with which to trace the index, one would expect the agreement features on the verb be copied into the NP-SUBJ node instead of leaving it completely empty (NP-Subj (-NONE- *). Arabic allows the subject to drop only if the verb is inflected as the inflection on the verb provides information on the subject. Copying these features from the verb into the subject node could be of great help in MT systems.

(23) (S (VP أَخلدَ *›aḫlda*::->axolad+a::eternalize/perpetuate/remain+he/it_[verb]

(NP-SBJ (-NONE- *))
(PP-CLR إِلَى *ʾilā*::<'ilaY::to/towards
(NP اَلنَوم *al-nawmi*::Al+nawom+i::the+sleep+[def.gen.])))
.::.::nogloss)
>axolad+a + <ilaY + Al+nawom+i
perpetuate + to + the+sleep
He retired to bed

4.5.3 The Noun Construct

While English marks genitive phrases such as "the cousin of the king" and "the king's cousin" by the explicit markers "of" and "'s", Arabic does not use any explicit marker for such construction. It joins the two nouns together which corresponds to "cousin king". Arab grammarians called this construction "iDaafa' إِضَافة *ʾiḍāft* which literally means 'addition'. They called the first term of the construct مضَافٌ *mḍāf* while the second term was called مُضَافٌ إِليهِ *muḍāfun ʾilyhi*. This construction exhibits interesting properties that distinguish it from other noun phrases. For example, The first term of such a construct is almost always indefinite. It governs the second element and assigns it genitive case. Like English, it is recursive. The whole construct acquires its gender and number from the first term while its definiteness is determined by the definiteness of the last term of the construct.

Although the noun construct is analyzed correctly in the ATB trees as a noun phrase, this construction is not marked as a possessive noun phrase or as a noun construct. The noun construct is very common in Arabic and categorizing it as a possessive NP or as an Idaafa or as a noun construct is necessary for correct translation it by machine translation systems.

4.5.4 Prepositions and Auxiliary Verbs

There is a list of "true" prepositions on page (33) of the guidelines. A true preposition like فِي *fiy* 'in' is not on the list nor is it found in the list of "nouns" formerly treated as prepositions on page 35. All Arab grammarians know that the particle اَلأَدَاة *al-ʾadāt*, is a "rag bag" category. The ATB categorizes auxiliaries and modals such as قَد *qad* as particles. This is not very informative. For example, قَد *qad* occurs only before verbs and has a specific meaning when it precedes an imperfect verb as in

قَد أَحصلُ عَلَى الوَظيفَةِ *qad ʾaḥṣlu ʿalā 'l-waẓyfati* 'I may get the job'

It has a different grammatical meaning when it is followed by a perfect verb as in قَد أَتَى *qad ʾatā* 'He has come'. In the later case, it is best categorized as a aspect marker. Thus, the morphological analyzer should have two tags for قَد *qad*: one as a modal meaning 'may'. This is the tag it should have when it comes before an imperfect verb. The other tag would mark it and its variant لَقَد *laqad* as a perfect aspect marker. This is its function when it appears before a perfect verb. More detailed analysis of functional words is needed.

```
(TOP
(S (CONJ wa-::وَ wa-::and )
(VP (PRT (NEG_PART -lam:: لَمْ lam:did_not ))
(IV3MS+IV+IVSUFF_DOD:J ya+tasan a+[null]::
يتسن ytsn]::he/it+
be_possible+[jus.] )
(PP-TMP (PREP EalaY::علَى ʿlā::on/above )
(NP (DET+NOUN+CASE_DEF_GEN Al+fawor+iاَلفَورِ al-fawr::
the+immediately+[def.gen.] )))
(NP-SBJ (NOUN+NSUFF_FEM_SG+CASE_DEF_NOM maEorif+ap+u:مَعرِفة marift:
: knowledge /acquaintance+[fem.sg.]+[def.nom.] )
(NP (NOUN+CASE_DEF_GEN sabab+i:سَبَب sabab::
reason/cause+[def.gen.] )
(NP (DET+NOUN+CASE_DEF_GEN Al+HAdiv+i::اَلحَادِث a-lḥādiṯ::
the+accident/mishap+[def.gen.] )))))))
```

وَلَم يَتَسَنَى عَلَى الفَورِ مَعرِفةِ سَبَب الحَادِث *walam yatasanā ʿalā 'l-fawri marifti sabab al-ḥādiṯ*

```
wa+lam + yatasan a+o + Ealay + Al+fawor+i + maEorifap+u + sabab+i +
Al+HAdiv+i
and+did not + be possible + on + the+immediately + knowledge + reason +
the+accident
It was not immediately possible to recognize the reason behind the
accident.
```

FIGURE 3 parse

.

4.5.5 Absence of X0 in NPs and ADJP

The guidelines states that "For ease of reading, however, most of the example trees in these guidelines will be shown without the part-of-speech/morphological tag nodes" (Maamouri et al., 2009, pg. 11). Although the absence of the morphological tags obscure the syntactic analysis, one would expect at least all heads of maximal projections would be marked. Categories such as VP, PrepP, NP, and AdjP are usually analyzed as maximal projections of their lexical heads i.e. X0 in the X-BAR notation (Jackendoff, 1977). X0 is usually the head of the maximal projection and every maximal projection must have a head and may have optional specifiers and/or complements. The analysis of Arabic sentences in the ATB guidelines shows verb phrases and prepositional phrases dominating verbs and prepositions, respectively. However, we don't see the NP or the AdjP as a projection of its head. An example from the ATB guidelines is given in figure 3.

In the above analysis we have the NP-SUBJ which has at least one noun construct "the reason of the accident". No constituent in this noun phrase is marked as head. We would expect that مَعرِفة *marift* 'knowledge' would be the head of the whole NP and

that سَبَب *sabab* would be the head of the lower NP.

Specifying the head is particularly important when analyzing the noun construct. The head of the gender and number of the head of the noun construct is percolated to the maximal projection. Consider the following two sentences:

(24) إِستَقبَلَ مُديرُ البَنكِ الضَيفَ *ʾistaqbala mudyru ʾl-banki ʾl-ḍayfa*
received-masc manager-masc the-bank the-guest
The bank manager received the guest.

(25) إِسَقبَلَت مُديرةُ البَنكِ الضَيفَ *ʾisaqbalat mudyrtu ʾl-banki ʾl-ḍayfa*
received-fem manager-fem the-bank the-guest
The bank manager received the guest.

The subject in (24) is the whole noun construct "the manager of the bank". The noun construct structure is [N [NP]]. The gender and number of the headhead of the construct becomes the gender and number of the whole construct. In (25) the gender of the head of the noun construct is feminine and becomes the gender of the whole noun construct. Thus, the verb agrees with the subject in gender and is feminine while the verb in (24) agrees in gender with the noun construct which is masculine. The point is, that it is very important in the noun construct to mark the head noun explicitly since the whole noun construct acquires its number and gender from the head of the construct and acquires its definiteness from the last term of the noun construct known in traditional grammar as مضَافٌ إِلَيهِ *mḍāfun ʾilayhi.*

4.5.6 Prepositional Attachment and Syntactic Boundaries

Attaching prepositions to the right node and identifying where a syntactic constituent ends are two of the most challenging problems in computational linguistics. One would expect that the ATB would provide enough training data that could help machine learning algorithms "learn" the right features so as to resolve these issues. Consider the following two sentences:

(26) أَعطَيتُ كِتَابًا لِعَلي *ʾaṭaytu kitāban liʿaly*
gave-I a-book to-Ali
I gave a book to Ali.

(27) قَرأَتُ كِتَابًا لِعَلي *qarʾatu kitāban liʿaly*
read-I a-book to-Alki
I read a book by Ali.

The phrase 'a book to Ali' is not one constituent as 'book' and 'Ali' here are two different entities. The preposition 'to' in (26) modifies the verb 'give'. The giving was to Ali. In contrast, the preposition in (27) modifies 'the book' and the phrase 'a book by Ali' is one constituent that could be an answer to the question 'which book did you read'. Now we expect that a parser generates two different parses of (26) and (27) with the preposition in (26) attached to the verb and the preposition in (27) attached to the noun phrase.

Passing sentences (26) and (27) to the Stanford Arabic parser which was trained on the ATB annotated corpus of MSA, we get the parse trees shown in (28) and (29).

(28) Parse of sentence (26)
أَعطَيتُ كِتَابًا لِعَلي *ʾaṭaytu kitāban liʿaly*
Tagging
أَعطَيتُ *ʾaṭaytu* /VBD
كِتَابًا *kitāban* /NN
لِعَلي *liʿaly* /JJ
Parse

(ROOT
(S
(VP (VBD أَعطَيتُ *ʾaṭaytu*
(NP (NN كِتَابًا *kitāban* (JJ لِعَلي *liʿaly*))))

The parse tree above shows the preposition modifying the NP and not the verb.
Parse of sentence (27)

(29) قَرأَتُ كِتَابًا لِعَلي *qarʾatu kitāban liʿaly*
Tagging
قرأَتُ *qrʾatu* /VBD
كِتَابًا *kitāban* /NN
لِعَلي *liʿaly*/JJ
Parse

(ROOT
(S
(VP (VBD قَرأَتُ *qarʾatu*)
(NP (NN)كِتَابًا *kitāban* (JJلِعَلي *liʿaly*)))))

The parse tree of sentence (27) is the correct one because it shows the preposition as modifying the NN. However, the structure in (27) is identical to that in (26) which signals that the parser should have detected the difference in the prepositional attachment in (26) and (27). Since the parser is trained on the annotation provided by the ATB, it is clear that it did not have enough data to be able to "learn" to distinguish between the two possible attachments. This is an area where the ATB could provide corpora focusing on the problematic areas in Arabic computational linguistics. Figures 4 and 5 below show the difference in structure of (28) and (29).

Another problem that faces Arabic computational linguists is finding where a constituent ends. It is usually easy to learn where a syntactic constituent begins. It usually has a head and the head usually occurs in the early part of a constituent in head-initial languages. It is difficult especially in Arabic, to define the point where a NP ends and the other begins. Consider the following sentence:

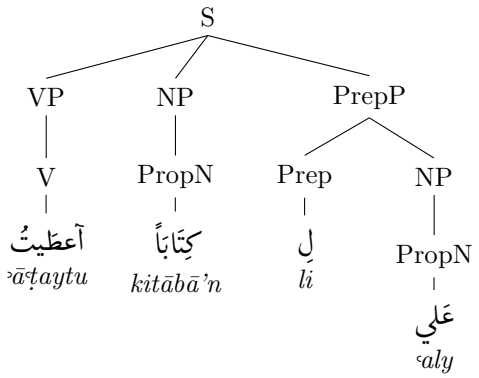

FIGURE 4 The prepositional phrase as an argument of the verb

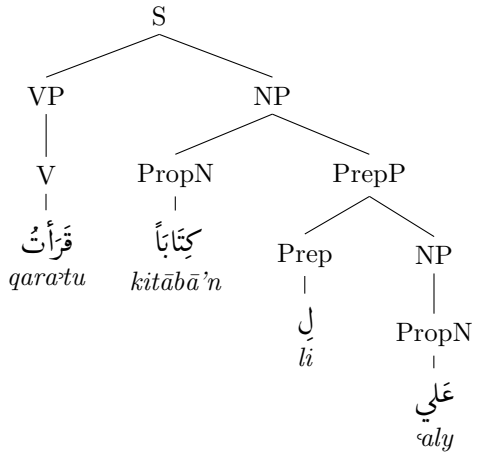

FIGURE 5 The prepositional phrase modifying the head noun

(30) قَابَلَت المُديرَةُ الضَيفَ *qābalat al-mudyratu 'l-ḍayfa*
met-she the-manager the-guest
The manager met the guest.

A speaker of Arabic would say this sentence consists of a verb followed by a subject and an object and is a regular VSO Arabic sentence. We passed the sentence to the Stanford Arabic parser. The following is the parse tree.

(31) قَابَلَت المُديرةُ الضَيفَ *qābalat al-mudyrtu 'l-ḍayfa*
Tagging
قَابَلَت *qābalat* /VBD
اَلمُديرَة *al-mudyrat* /DTNN
اَلضَيف *al-ḍayf* /DTNN
Parse

(ROOT
(S
(VP (VBD قَابَلَت *qābalat*)
(NP
(NP (DTNN اَلمُديرَةُ *al-mudyratu*))
(NP (DTNN اَلضَيفَ *al-ḍayfa*))))))

In this parse tree, there are only two nodes that are immediately dominated by the VP: they are the VBD and the higher NP. The higher NP node immediately dominates two NPs which are the subject and the direct object of the sentence. Assigning the subject and direct object to one node equates them as one argument. One would expect the direct object to have a sister node with the verb equal to that of the subject. The problem here is to find out when the two noun phrases form one constituent and thus can have a higher NP node dominating them, or when they are two different arguments and should not have a common node immediately dominant. In order for a parser to learn to differentiate the two structures, a large annotated corpora focusing on the problematic areas in Arabic computational linguistics would be extremely beneficial.

5 Conclusion

We have looked at the Arabic language from historical, cultural, strategic and linguistic view points. The stability of the Arabic language, its inseparable ties to Islam, and its interesting linguistic properties raise many interesting questions for research. Furthermore, an understanding of its history, culture and linguistic properties is key to productive cultural dialog with the West. Interest in the Arabic language and culture surged in the last few years and as a result, several state-of-the-art computational tools have been developed using machine learning and engineering knowledge. Such efforts could benefit from a fresh look at the Arabic language and its structure. We presented some of the contributions of traditional Arabic grammarians and pointed out that the grammar they developed- which works well for Classical Arabic, does not provide sufficient rules for

computational treatments of MSA texts. Finally, we talked about computational Arabic morphology and pointed to the absence of descriptions of the morphology-syntax interface. We looked at the recent Arabic treebank guidelines and pointed out some possible improvements.

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